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Quasicoherent sheaves

on affine toric stacks III

Last time Stated the Fourier transform, background on (co)monadic descent

Today A bit on (co)monads Proofs that the Fourier transform is an equivalence and that

$$\mathrm{QCoh}(X_\sigma/T) \cong \mathrm{Fun}(\mathcal{O}(\sigma)^\mathrm{op}, \mathrm{Sp})$$

> On the next page, we'll recall the key tool we introduced last time

Thm (Beck for 1-categories, Lurie for ∞ -categories) Let $e^*: \Delta_+ \rightarrow \text{Cat}_\infty$ be an augmented cosimplicial ∞ -category write $G: e^{-1} \rightarrow e^0$ for the augmentation Assume

(1) G preserves limits of cosimplicial diagrams that become split after applying G .

(2) For each map $\alpha: [m] \rightarrow [n]$ in Δ_+ , the commutative square

$$\begin{array}{ccc} e^m & \xrightarrow{d^0} & e^{m+1} \\ \alpha_* \downarrow & \curvearrowright R_m & \downarrow ([0] + \alpha)_* \\ e^n & \xrightarrow{d^0} & e^{n+1} \\ & \curvearrowleft R_n & \end{array}$$

is horizontally right adjointable.

$$\alpha_* R_m \xrightarrow{\sim} R_n \circ ([0] + \alpha)_*$$

Then $e^{-1} \rightarrow \lim_{[n] \in \Delta} e^n$ admits a fully faithful right adjoint.

If G is also conservative, then this functor is an equivalence.

Digression: (co)monads

- > This is to explain where the previous theorem comes from, but we don't explicitly need it.

Def. Let $\mathcal{C}$ be an ∞ -category. Then

$$\mathrm{End}(\mathcal{C}) = \mathrm{Fun}(\mathcal{C}, \mathcal{C})$$

has a natural monoidal structure given by composition, with unit $\mathrm{id}_{\mathcal{C}}$. A monad on $\mathcal{C}$ is an algebra T in $\mathrm{End}(\mathcal{C})$. A comonad on $\mathcal{C}$ is a coalgebra in $\mathrm{End}(\mathcal{C})$.

Example. Let $L: \mathcal{C} \rightleftarrows \mathcal{D}, R$ be an adjunction. Then $LR: \mathcal{C} \rightarrow \mathcal{C}$ refines to a monad on $\mathcal{C}$ with unit $\mathrm{id}_{\mathcal{C}} \rightarrow LR$ the unit of the adjunction and multiplication

$$LR \circ LR \simeq L \circ RL \circ R \xrightarrow{L \circ \mathrm{counit} \circ R} LR.$$

Dually, $RL: \mathcal{D} \rightarrow \mathcal{D}$ refines to a comonad on $\mathcal{D}$ with counit $RL \rightarrow \mathrm{id}_{\mathcal{D}}$ the adjunction counit and comultiplication

$$R \circ \text{unit} \circ L : RL \longrightarrow RLRL$$

Informal Definition. Let $T : \mathcal{C} \longrightarrow \mathcal{C}$ be a monad. One can define an ∞ -category $\text{Mod}_T(\mathcal{C})$ of (left) T -modules in $\mathcal{C}$ with objects informally given by pairs $(c \in \mathcal{C}, a : T(c) \rightarrow c)$ so that the diagrams

$$\begin{array}{ccc} c & \xrightarrow{\text{unit}} & T(c) \\ & \searrow & \downarrow a \\ & & c \end{array} \qquad \begin{array}{ccc} T^2(c) & \xrightarrow{\text{mult}} & T(c) \\ T(u) \downarrow & & \downarrow a \\ T(c) & \xrightarrow{a} & c \end{array}$$

commute + all higher coherences

- > There is a forgetful functor $\text{Mod}_T(\mathcal{C}) \longrightarrow \mathcal{C}$ sending $(c, a) \mapsto c$
- > It has a left adjoint free functor $c \mapsto (T(c), \text{mult} : T^2(c) \rightarrow T(c))$

Upshot. The monad $T: \mathcal{C} \rightarrow \mathcal{C}$ is isomorphic to

$$\mathcal{C} \xrightarrow{\text{free}} \text{Mod}_T(\mathcal{C}) \xrightarrow{\text{forget}} \mathcal{C}$$

Def. A right adjoint $R: \mathcal{D} \rightarrow \mathcal{C}$ is monadic if there exists a monad $T: \mathcal{C} \rightarrow \mathcal{C}$ and an equivalence over $\mathcal{C}$

$$\begin{array}{ccc} \mathcal{D} & \xrightarrow{\sim} & \text{Mod}_T(\mathcal{C}) \\ & \searrow R & \swarrow \text{forget} \\ & \mathcal{C} & \end{array}$$

> Here's what's behind the key tool.

Monadicity Thm (Barr-Beck-Lurie). Let $\mathcal{C}$ be an ∞ -category and $G: \mathcal{D} \rightarrow \mathcal{C}$ a left adjoint. TFAE:

(1) G is comonadic.

(2) G is conservative and preserves limits of cosimplicial diagrams that become split after applying G .

The Fourier Transform

Thm. Let $M \in \text{Grp}(\text{Set})$. The Sym. mon. functor

$$\Phi_M: \text{Fun}(M, \text{Sp}) \longrightarrow \text{QCoh}(\text{BD}_M) \simeq \lim_{[n] \in \Delta} \text{QCoh}(* \times D_M^{\times n})$$

is an equivalence

Obs. If $F: \mathcal{C} \rightarrow \mathcal{D}$ is an exact functor between stable ∞ -categories, then F is conservative iff $F(c) = 0$ implies $c = 0$.

Proof

We apply the key tool to the diagram induced by Φ_M

$$\text{Fun}(M, \text{Sp}) \xrightarrow{G} \text{QCoh}(*) \rightrightarrows \text{QCoh}(* \times D_M) \rightrightarrows \dots$$

$\downarrow \text{SI}$ $\downarrow \text{SI}$
 Sp $\text{Mod}_{S(M)}$

> unwinding the definition of Φ_M shows that

$$G((X_m)_{m \in M}) := \bigoplus_{m \in M} X_m$$

Step 1. G is conservative

Obs. If $\mathcal{C}$ is a pointed ∞ -category and $X \in \mathcal{C}$ is a retract of 0, then $X = 0$.

Obs. In any stable ∞ -category, given objects $(X_i)_{i \in I}$, each X_j is a retract of $\bigoplus_{i \in I} X_i$. Indeed,

$$X \xrightarrow{\text{coprod inc}} \bigoplus_{i \in I} X_i \simeq X_j \oplus \bigoplus_{i \in I \setminus \{j\}} X_i \xrightarrow{\text{pr}_1} X_j$$

is the identity.

> So if $G((X_m)_{m \in M}) = \bigoplus_{m \in M} X_m$ is zero, then each X_m is zero.

Step 2. G preserves limits of G -split cosimplicial diagrams.

In fact, we'll prove:

Claim. Let $X^\bullet: \Delta \rightarrow \text{Fun}(M, \text{Sp})$ be a cosimplicial diagram with limit X^{-1} , and regard $X^{-1} \rightarrow X^\bullet$ as an augmented cosimplicial diagram $\bar{X}^\bullet$. If $G(\bar{X}^\bullet)$ is split, then $\bar{X}^\bullet$ is split, hence a limit diagram.

- > Write $\bar{X}^\bullet = (\bar{X}_m^\bullet)_{m \in M}$. By assumption, $\bigoplus_{m \in M} \bar{X}_m^\bullet$ is split.
- > In particular, each $\bar{X}_m^\bullet$ is a retract of a split augmented cosimplicial object.

Easy fact. In any ∞ -category, Split augmented cosimplicial diagrams are closed under retracts.

- > So actually each $\bar{X}_m^\bullet$ is split.
- > Thus the M -indexed collection of diagrams $\bar{X}^\bullet = (\bar{X}_m^\bullet)_{m \in M}$ is also split.

Step 3. Adjointability

General Lemma. Let $\mathcal{C}$ be a symmetric monoidal ∞ -category with geometric realizations, such that $\otimes$ preserves geometric realization in each variable.

Let

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

be a pushout square in $\mathbf{Catg}(\mathcal{C})$.

Then the induced square of forgetful functors

$$\begin{array}{ccc}
 \text{Mod}_A(e) & \xrightarrow{B \otimes_A (-)} & \text{Mod}_B(e) \\
 \downarrow C \otimes_A (-) & & \downarrow D \otimes_B (-) \\
 \text{Mod}_C(e) & \xrightarrow{D \otimes_C (-)} & \text{Mod}_D(e)
 \end{array}$$

is horizontally right adjointable.

Why? This is actually easy. It says that for $M \in \text{Mod}_B(e)$,

$$C \otimes_A M \simeq D \otimes_B M$$

forget to an A -mod

as C -modules. Indeed, $D \simeq C \otimes_A B$, so

$$D \otimes_B M \simeq C \otimes_A B \otimes_B M \simeq C \otimes_A M$$

> Let $\alpha: [m] \rightarrow [n]$ be in $\mathbb{A}_+$ with $m \neq -1$. Consider the comm diagram of affine stacks

$$\begin{array}{ccc}
 * \times D_M^{x(n+1)} & \xrightarrow{d_0} & * \times D_M^{x n} \\
 \downarrow ([0]_* \alpha)^* & & \downarrow \alpha^* \\
 * \times D_M \times D_M^{x m} \simeq * \times D_M^{x(m+1)} & \xrightarrow{d_0} & * \times D_M^{x m} \\
 \downarrow \text{id} \times \text{id} \times 0 & & \downarrow \\
 * \times D_M \times * \simeq * \times D_M & \longrightarrow & *
 \end{array}$$

Observe that the bottom and large outer squares are pullbacks. So the top square is a pullback.

> The adjointability result about module ∞ -categories proves the desired adjointability result for α .

> Let $\alpha [-1] \rightarrow [n]$ be in Δ_+ . Then α factors as

$$[-1] \xrightarrow{\text{unique}} [0] \xrightarrow{\beta} [n].$$

Since right adjointable squares compose, and we've proven the claim for β , it suffices to prove the claim when α is the unique map $[-1] \rightarrow [0]$.

> Let $p: M \rightarrow *$ and $\pi: D_M \rightarrow *$ be the unique maps unwinding definitions, we need to show that the square

$$\begin{array}{ccc} \text{Fun}(M, \mathcal{S}p) & \xrightarrow{G} & \mathcal{S}p \\ G \downarrow & & \downarrow d^1 \\ \mathcal{S}p & \xrightarrow{d^0} & \text{QCoh}(D_M) \end{array}$$

is horizontally right adjointable.

Note that the homotopy $d^0 G \simeq d^1 G$ filling this square is nontrivial.

> Unwinding definitions, this square is equivalent to

$$\begin{array}{ccccc}
 \text{Fun}(M, S_p) & \xrightarrow{p_{\#} = \text{colim}} & S_p & \simeq & \text{Mod}_S \\
 p_{\#} \downarrow & & \downarrow \pi^* & & \downarrow S\langle M \rangle \otimes_S (-) \\
 S_p & \xrightarrow{\pi^*} & \text{QCoh}(D_M) \simeq & \text{Mod}_{S\langle M \rangle} &
 \end{array}$$

but filled with a non-identity homotopy $\pi^* p_{\#} \simeq \pi^* p_{\#}$

- We need to see that

$$\text{Ex } p_{\#} p^* \xrightarrow{\text{unit}} \pi_* \pi^* p_{\#} p^* \xrightarrow{\sim} \pi_* \pi^* p_{\#} p^* \xrightarrow{\text{counit}} \pi_* \pi^*$$

is an equivalence

- $p_{\#}$, π^* , and the constant functor p^* all preserve colimits. Since π is affine π_* is also a left adjoint with right adj

$$\text{Hom}_{S_p}(S\langle M \rangle, -)$$

- So by the universal property of Spectra, it suffices to check that this map is an equivalence when evaluated on the sphere S .

> unwinding definitions, we see that $Ex(S)$ is given by

$$\bigoplus_M S \xrightarrow{\bigoplus_M \text{unit}} \bigoplus_M S\langle M \rangle \xrightarrow{\bigoplus_{m \in M} \text{mult by } m} \bigoplus_{m \in M} S\langle M \rangle \xrightarrow{\text{Sum}} S\langle M \rangle_{SI} \oplus S_M$$

The composite is $m \cdot S \rightarrow S\langle M \rangle$ on each summand, which is an equivalence because M is a group. $\square$

Proof that Φ_σ is an equivalence

Situation we're in. For a lattice N and cone $\sigma \in N_{\mathbb{R}}$, we have a commutative square

$$\begin{array}{ccc}
 \text{Fun}(M, \mathcal{S}_p) & \xrightarrow{P_{\sigma, \#}} & \text{Fun}(\Theta(\sigma)^p, \mathcal{S}_p) \\
 \Phi_M \downarrow \wr & & \downarrow \Phi_\sigma \\
 \text{QCoh}(BT) & \xrightarrow{\pi_\sigma^*} & \text{QCoh}(X_\sigma/T)
 \end{array}$$

(*)

Fact. Since $\pi: X_\sigma \rightarrow *$ is affine, $\pi_\sigma: X_\sigma/T \rightarrow BT$ is an affine morphism of stacks. So $\pi_{\sigma, *}$ is conservative and admits a right adjoint.

Lemma. The square (*) is horizontally right adjointable.

> The proof isn't difficult, but is technical. Ultimately, it reduces to the closure of adjointable squares under limits [HA, Corollary 4.7.4.18].

Upshot We're in the situation that Φ_σ fits into a commutative square

$$\begin{array}{ccc}
 \text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}) & \xrightarrow{\Phi_\sigma} & \text{QCoh}(X_\sigma/T) \\
 \downarrow p_\sigma^* & & \downarrow \pi_{\sigma,*} \\
 \text{Fun}(M, \text{Sp}) & \xrightarrow[\Phi_M]{\sim} & \text{QCoh}(BT)
 \end{array}$$

(**)

> Here's the key categorical result we need

Comparison Lemma for monadic adjunctions. Let

$$\begin{array}{ccc}
 \mathcal{C} & \xrightarrow{U} & \mathcal{C}' \\
 & \searrow R & \swarrow R' \\
 & \mathcal{D} &
 \end{array}$$

be a commutative triangle of ∞ -categories.

Assume

(1) R and R' admit left adjoints L and L'

(2) R and R' are both conservative and preserve colimits

of simplicial diagrams that become split after applying these functors

(3) For each $d \in D$, the map

$$\alpha_d: L'(d) \longrightarrow UL(d)$$

adjoint to the map

$$\text{unit } d \longrightarrow RL(d) \simeq R'(UL(d))$$

is an equivalence

Then U is an equivalence

Thm. $\Phi_\sigma: \text{Fun}(\Theta(\sigma)^{\text{op}}, S_p) \longrightarrow \text{QCoh}(X_\sigma/T)$ is an equivalence

Proof.

In light of the diagram (**), we need to check (1) - (3) above

(1) Clear

by conservativity
 $\iff R'(d_d): R'L'(d) \rightarrow RL(d)$
 is an equivalence

(2) As noted previously, p_σ^* and $\pi_{\sigma,*}$ are both left adjoints, hence preserve colimits. Since π_σ is affine, $\pi_{\sigma,*}$ is conservative.

For the conservativity of p_σ^* , note that

$$\begin{array}{ccc} p_\sigma: M & \longrightarrow & \mathcal{O}(\sigma) \\ m & \longmapsto & m + \sigma^\vee \end{array}$$

is surjective. So the claim follows from the fact that the pullback along an essentially surjective functor is conservative.

(3) This says that the specified map

$$\pi_\sigma^* \longrightarrow \Phi_\sigma p_{\sigma,\#} \Phi_M^{-1}$$

is an equivalence. This follows from the commutativity of (*): equivalently,

$$\pi_\sigma^* \Phi_M \longrightarrow \Phi_\sigma p_{\sigma,\#} \text{ is an equivalence. } \square$$